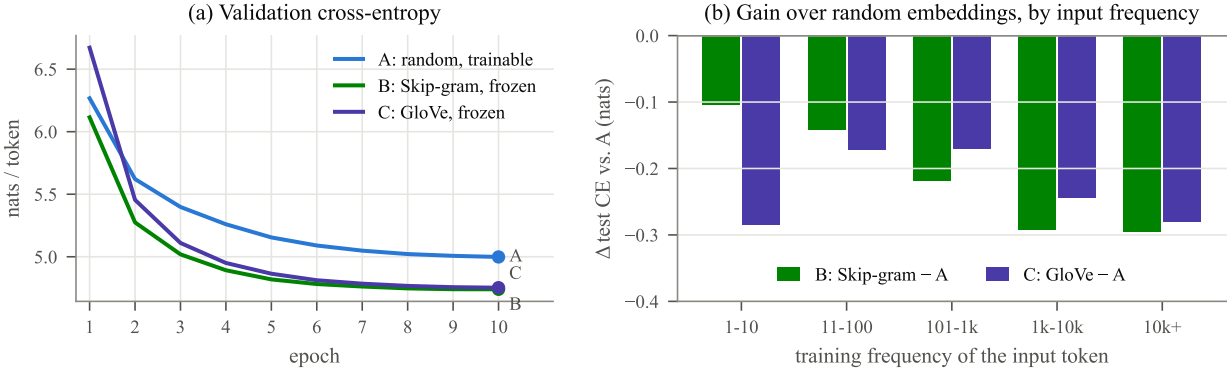
(a) Validation cross-entropy
A: random, trainable
B: Skip-gram, frozen
C: GloVe, frozen
nats / token
6.5
6.0
5.5
5.0
A
C
B
1
2
3
4
5
6
7
8
9
10
epoch
(b) Gain over random embeddings, by input frequency
Δ test CE vs. A (nats)
0.0
−0.1
−0.2
−0.3
−0.4
B: Skip-gram − A
C: GloVe − A
1-10
11-100
101-1k
1k-10k
10k+
training frequency of the input token